

THE COMPLEX CROSS-POLYTOPE $B_1(\mathbb{C}^2)$: REAL FRACTIONAL ILLUMINATION NUMBER 2^2 , AND NOT A COMPLEX-LINEAR IMAGE OF THE POLYDISC

ABSTRACT. Rotem, Schejter and Slomka conjectured that every complex convex body $K \subseteq \mathbb{C}^n$ satisfies $\text{ill}^*(K) \leq 2^n$, with equality if and only if K is a linear image of the polydisc D^n . Reading ill^* as the ordinary real fractional illumination number of K regarded as a convex body in $\mathbb{R}^{2n}$ — a reading we do not check against the conjecture’s own wording of ill^* — the complex cross-polytope $B_1(\mathbb{C}^2) = \{z : |z_1| + |z_2| \leq 1\}$ satisfies $\text{ill}^*(B_1(\mathbb{C}^2)) = 4 = 2^2$, and it is not a complex-linear image of D^2 . So under that reading of ill^* , and with “linear image” read as complex-linear, the equality clause fails at $n = 2$. The bound $\text{ill}^*(K) \leq 2^n$ itself is not contradicted — the witness attains it with equality — and the equality clause is untouched for $n \geq 3$, because $\text{ill}^*(B_1(\mathbb{C}^n)) = 2n < 2^n$ there.

1. THE CONJECTURE

Let $D \subseteq \mathbb{C}$ be the closed unit disc and $D^n = D \times \cdots \times D$ the polydisc. Following Rotem, Schejter and Slomka [1], a *complex convex body* in $\mathbb{C}^n$ is the unit ball of a norm on $\mathbb{C}^n$; equivalently it is a centrally symmetric convex body $K \subseteq \mathbb{R}^{2n}$ with $x \in K \Rightarrow e^{i\theta}x \in K$ for all $\theta \in \mathbb{R}$. A direction v *illuminates* a boundary point x of a convex body K if $x + tv \in \text{Int } K$ for some $t > 0$; writing $A_K(x)$ for the set of directions illuminating x , a non-negative Borel measure μ *illuminates* K if $\mu(A_K(x)) \geq 1$ for every $x \in K$, and

$$\text{ill}^*(K) = \inf\{\mu(\mathbb{R}^{2n}) : \mu \text{ illuminates } K\}$$

is the *fractional illumination number* of K , introduced by Naszódi [2]. Throughout, ill^* of a complex body is read as the ordinary real fractional illumination number of K regarded as a convex body in $\mathbb{R}^{2n}$: real directions, and no circle invariance imposed on μ . We do not reproduce [1]’s own wording of that definition here, and every value below is asserted for this reading only; under a reading that restricted μ to circle-invariant measures the numbers could differ. The measure we exhibit below is supported on unit vectors, so it is admissible whether or not non-unit vectors are allowed. Everything below is therefore conditional on [1]’s ill^* agreeing with the reading just fixed; we have not established that it does.

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The statement we address is Conjecture 1.2 of [1]. In the version of record it reads, verbatim:

Conjecture 1.2. Given a complex convex body $K \subseteq \mathbb{C}^n$, we have $\text{ill}^*(K) \leq 2^n$. Moreover, equality holds if and only if K is a linear image of D^n .

We quote the published wording; the companion Conjecture 1.1 of [1] concerns the classical illumination number ill and is *not* addressed here.

Two theorems of [1] delimit what was known: $\text{ill}^*(D^n) = 2^n$ for all $n \geq 1$, and every full-dimensional complex zonoid $Z \subseteq \mathbb{C}^n$ that is not a linear image of D^n has $\text{ill}^*(Z) < \text{ill}^*(D^n)$. By the second of those theorems, any counterexample to the equality clause must lie outside the complex zonoids; Section 5 shows that our witness, the polar of the polydisc, does.

2. THE RESULT

Theorem 1. Let $K = B_1(\mathbb{C}^2) = \{z \in \mathbb{C}^2 : |z_1| + |z_2| \leq 1\}$. With ill^* read as in Section 1,

$$\text{ill}^*(B_1(\mathbb{C}^2)) = 4 = 2^2,$$

and $B_1(\mathbb{C}^2)$ is not a complex-linear image of D^2 . Consequently, if ill^* in Conjecture 1.2 of [1] is that same number and its “linear image” is meant complex-linearly, the equality clause fails at $n = 2$: the bodies in $\mathbb{C}^2$ attaining $\text{ill}^* = 2^2$ include at least the two $GL_2(\mathbb{C})$ -orbits of D^2 and of $B_1(\mathbb{C}^2)$.

We state at once what is *not* claimed, since the conjecture is a conjunction and only one conjunct falls.

- The two readings fixed in Section 1 are not shown to be [1]’s. We have not reproduced their definition of ill^* , and we settle only complex-linear images of D^2 ; whether $B_1(\mathbb{C}^2)$ is a real-linear image of D^2 is not decided here.
- The bound $\text{ill}^*(K) \leq 2^n$ is *not* contradicted. The witness satisfies it with equality, and we have no evidence against it; the first conjunct of Conjecture 1.2 remains open.
- Nothing is claimed for $n \geq 3$. By Proposition 4 below, $\text{ill}^*(B_1(\mathbb{C}^n)) = 2n < 2^n$ for $n \geq 3$, so this witness is silent there and the equality clause may well hold as written for every $n \geq 3$.
- Even at $n = 2$ we show only that the equality set is strictly larger than the orbit of D^2 . Whether $\{D^2, B_1(\mathbb{C}^2)\}$ exhausts it is open.
- Conjecture 1.1 of [1], on the classical illumination number, is untouched; we make no claim about it.

3. THE WITNESS

Throughout, $K = B_1(\mathbb{C}^2)$, and we use the real pairing $\langle u, z \rangle = \text{Re}(u_1 \bar{z}_1 + u_2 \bar{z}_2)$ on $\mathbb{C}^2 \cong \mathbb{R}^4$.

(a) K is a complex convex body. $\|z\| := |z_1| + |z_2|$ is a norm on $\mathbb{C}^2$ and K is its unit ball; K is full-dimensional (real dimension 4) and $\|e^{i\theta}z\| = \|z\|$, so K is circle-invariant. Moreover

$$K = \text{conv}(De_1 \cup De_2) = (D^2)^\circ,$$

the first equality because $z = |z_1|(\frac{z_1}{|z_1|}, 0) + |z_2|(0, \frac{z_2}{|z_2|}) + (1 - \|z\|) \cdot 0$ is a convex combination for every $z \in K$.

(b) Support function.

$$h_K(u) = \sup_{|z_1|+|z_2|\leq 1} \langle u, z \rangle = \sup_{0\leq s\leq 1} (|u_1|s + |u_2|(1-s)) = \max(|u_1|, |u_2|),$$

the complex ℓ^∞ norm — the polar situation to D^2 , whose support function is $|u_1| + |u_2|$.

(c) The boundary, exhaustively. ∂K consists of exactly three classes:

(i) $|z_1| = 1, z_2 = 0$; (ii) $z_1 = 0, |z_2| = 1$; (iii) *mixed* points, with both coordinates nonzero and $|z_1| + |z_2| = 1$.

(d) Extreme points: two disjoint circles.

$$\text{ext}(K) = C_1 \cup C_2, \quad C_1 = \{(e^{i\alpha}, 0)\}, \quad C_2 = \{(0, e^{i\beta})\}.$$

Indeed $\text{ext}(\text{conv}(A \cup B)) \subseteq \text{ext}(A) \cup \text{ext}(B)$ with $A = De_1, B = De_2$ gives the inclusion, and each such point is exposed: on K , $\text{Re}(e^{-i\alpha}z_1) \leq |z_1| \leq |z_1| + |z_2| \leq 1$ with equality if and only if $z_1 = e^{i\alpha}$ and $z_2 = 0$.

(e) The illuminating measure, in closed form.

(1)

$$\mu = 2 \cdot \text{Unif}\{(e^{i\varphi}, 0) : \varphi \in [0, 2\pi)\} + 2 \cdot \text{Unif}\{(0, e^{i\psi}) : \psi \in [0, 2\pi)\}, \quad \mu(\mathbb{R}^4) = 4.$$

It is supported on unit vectors.

(f) Illumination criteria. Since $\|(e^{i\alpha}, 0) + tv\| = |e^{i\alpha} + tv_1| + t|v_2| = 1 + t(\text{Re}(e^{-i\alpha}v_1) + |v_2|) + O(t^2)$,

$$v \text{ illuminates } (e^{i\alpha}, 0) \iff \text{Re}(e^{-i\alpha}v_1) + |v_2| < 0,$$

$$(2) \quad v \text{ illuminates } (0, e^{i\beta}) \iff \text{Re}(e^{-i\beta}v_2) + |v_1| < 0,$$

$$v \text{ illuminates a mixed } x \iff \text{Re}(e^{-i\arg x_1}v_1) + \text{Re}(e^{-i\arg x_2}v_2) < 0.$$

The same criteria follow from the normal-cone description used in [1] (namely, v illuminates x iff $\langle u, v \rangle < 0$ for every nonzero outer normal u at x , cf. [3, (2.2)]): at $x = (e^{i\alpha}, 0)$ the normal cone is $\{(re^{i\alpha}, u_2) : r \geq |u_2|\}$, and at a mixed x it is the single ray spanned by $(e^{i\arg x_1}, e^{i\arg x_2})$.

4. PROOF THAT $\text{ill}^*(B_1(\mathbb{C}^2)) = 4$

Lower bound $\text{ill}^(K) \geq 4$. Step 1: the two extreme circles are illumination-disjoint.* If v illuminates some point of C_1 then, by (2), $\text{Re}(e^{-i\alpha}v_1) < -|v_2|$, whence $|v_1| \geq -\text{Re}(e^{-i\alpha}v_1) > |v_2|$. Symmetrically, illuminating a point of C_2 forces $|v_2| > |v_1|$. Hence $A_K(C_1) \subseteq V_1 := \{|v_1| > |v_2|\}$ and $A_K(C_2) \subseteq V_2 := \{|v_2| > |v_1|\}$, and $V_1 \cap V_2 = \emptyset$: no single direction serves both circles.

Step 2: exact arc length. Fix $v \in V_1$ and write $v_1 = re^{i\varphi}$ with $r > 0$ and $c = |v_2|/r \in [0, 1)$. By (2), v illuminates $x_\alpha := (e^{i\alpha}, 0)$ if and only if $\cos(\varphi - \alpha) < -c$, an arc of α whose endpoints satisfy $\cos(\varphi - \alpha) = -c$ and whose length is exactly

$$2\pi - 2 \arccos(-c) = 2 \arccos(c) \leq \pi,$$

with length π only when $v_2 = 0$. (If $r = 0$ the condition reads $|v_2| < 0$, which never holds.)

Step 3: averaging. Let μ illuminate K . Integrating the constraint $\mu(A_K(x_\alpha)) \geq 1$ over $\alpha \in [0, 2\pi)$ and exchanging the order of integration (Tonelli; all integrands are non-negative),

$$2\pi \leq \int_0^{2\pi} \mu(A_K(x_\alpha)) d\alpha = \int_{V_1} 2 \arccos(|v_2|/|v_1|) d\mu \leq \pi \mu(V_1),$$

so $\mu(V_1) \geq 2$, and identically $\mu(V_2) \geq 2$. The two sets are disjoint, so $\mu(\mathbb{R}^4) \geq 4$. $\square$

Upper bound $\text{ill}^(K) \leq 4$.* Take μ as in (1) and check the constraint on every boundary class directly, so that no extreme-point reduction is needed.

- $x = (e^{i\alpha}, 0)$: by (2), $(e^{i\varphi}, 0)$ illuminates x iff $\cos(\varphi - \alpha) < 0$, an open half-circle, of measure exactly 1 under the mass-2 uniform measure on that circle; and $(0, e^{i\psi})$ never does, since the criterion would read $|e^{i\psi}| = 1 < 0$. Total $\mu(A_K(x)) = 1$.
- $x = (0, e^{i\beta})$: symmetric, total 1.
- x mixed: each axis circle contributes its open half-circle, total $1+1 = 2$.
- $x \in \text{Int } K$: every direction illuminates x , total 4.

Hence μ illuminates K and $\text{ill}^*(K) \leq \mu(\mathbb{R}^4) = 4$. $\square$

Combining the two bounds gives $\text{ill}^*(B_1(\mathbb{C}^2)) = 4 = 2^2$.

5. $B_1(\mathbb{C}^2)$ IS NOT A COMPLEX ZONOID, HENCE NOT A COMPLEX-LINEAR IMAGE OF D^2

A complex-linear image of D^2 is a 2-generator complex zonotope, since $T(D^2) = T(De_1 + De_2) = DT(e_1) + DT(e_2)$; in particular it is a complex zonoid. So Theorem 1's second assertion follows from:

Proposition 2. $B := B_1(\mathbb{C}^2)$ is not a complex zonoid.

Proof: the mean-value obstruction. Recall from [1] that a complex zonoid $Z \subseteq \mathbb{C}^n$ admits a non-negative Borel measure ν on the unit sphere S of $\mathbb{C}^n$ with $h_Z(\theta) = \int_S |\langle x, \theta \rangle_{\mathbb{C}}| d\nu(x)$. Suppose such a ν existed for B , whose support function is $h_B(\theta) = \max(|\theta_1|, |\theta_2|)$. Testing $\theta = (1, 0)$ and $\theta = (0, 1)$ gives

$$(3) \quad \int_S |x_1| d\nu = 1, \quad \int_S |x_2| d\nu = 1,$$

and testing $\theta = (1, e^{it})$ for every real t gives $\int_S |x_1 + e^{it}x_2| d\nu = 1$, since $h_B = \max(1, 1) = 1$. Averaging the last identity over $t \in [0, 2\pi)$ and exchanging the order of integration (Tonelli again),

$$\int_S M(|x_1|, |x_2|) d\nu = 1, \quad M(a, b) := \frac{1}{2\pi} \int_0^{2\pi} |a + e^{it}b| dt.$$

The map $w \mapsto a + wb$ is holomorphic on the unit disc, so $\frac{1}{2\pi} \int_0^{2\pi} (a + e^{it}b) dt = a$ and hence $M(a, b) \geq |a|$; writing $a + e^{it}b = e^{it}(b + e^{-it}a)$ gives $M(a, b) \geq |b|$ as well. Therefore $M(a, b) \geq \max(a, b) \geq (a + b)/2$ for $a, b \geq 0$, and with (3)

$$1 = \int_S M d\nu \geq \int_S \frac{|x_1| + |x_2|}{2} d\nu = \frac{1 + 1}{2} = 1.$$

Equality is forced, so $M(|x_1|, |x_2|) = (|x_1| + |x_2|)/2$ for ν -almost every x . But this is impossible on S , and the following three cases are exhaustive there:

- if $|x_1|, |x_2| > 0$ with $|x_1| \neq |x_2|$, no property of M beyond $M \geq \max$ is needed: the second inequality of $M \geq \max(a, b) \geq (a + b)/2$ is already strict, since $\max(a, b) > (a + b)/2$ whenever $a \neq b$;
- if $|x_1| = |x_2| = r > 0$ then $\max(a, b) = (a + b)/2$, so this is the one case in which the *value* of M must decide it, and it does: $M = r \cdot \frac{1}{2\pi} \int_0^{2\pi} 2|\cos(t/2)| dt = \frac{4r}{\pi} = 1.2732 \dots r > r = \max(r, r)$, using $|a + e^{it}a| = 2a|\cos(t/2)|$ and $\int_0^{2\pi} |\cos(t/2)| dt = 4$;
- if $x_2 = 0$ then $M = |x_1|$ while $(|x_1| + 0)/2 = |x_1|/2$, forcing $x_1 = 0$, impossible on S ; symmetrically if $x_1 = 0$.

In every case $M(|x_1|, |x_2|) > (|x_1| + |x_2|)/2$ strictly, so ν charges no point of S . Hence $\nu = 0$, contradicting (3). $\square$

Remark 3. No theorem of [1] is contradicted: the theorem that a full-dimensional complex zonoid other than a linear image of D^n has ill^* *strictly* below $\text{ill}^*(D^n)$ does not apply to our witness, precisely because of Proposition 2.

6. THE WITNESS IN $\mathbb{C}^n$

Proposition 4. $\text{ill}^*(B_1(\mathbb{C}^n)) = 2n$ for every $n \geq 1$.

Proof. As in (2), v illuminates $e^{i\alpha}e_i$ if and only if $\text{Re}(e^{-i\alpha}v_i) + \sum_{j \neq i} |v_j| < 0$, which forces $|v_i| > \sum_{j \neq i} |v_j|$; adding two such inequalities for $i \neq k$ yields a contradiction, so the n extreme circles are pairwise illumination-disjoint. The argument of §4 then gives mass at least 2 per circle, and mass exactly 2 per circle suffices: a boundary point whose set of nonzero coordinates is S receives exactly $|S| \geq 1$. $\square$

Now $2n = 2^n$ holds only at $n = 1$ and $n = 2$, and $2n < 2^n$ for $n \geq 3$. So this witness attains the conjectured extremal value only in $\mathbb{C}^2$ and lies strictly below it for $n \geq 3$, where the equality clause is untouched by it. We have

not carried out a systematic priority check for the formula $\text{ill}^*(B_1(\mathbb{C}^n)) = 2n$ and make no claim that it is new; it may well be recorded elsewhere.

7. THE ACCOMPANYING PROGRAM

All proofs above are closed form. The program `verify.py` in this folder, with its recorded run `verify.output.txt`, is a sampled consistency check in exact rational arithmetic: it parses the body, the support function and the measure (1), tests the criteria (2) against the definition of illumination (membership of $x + tv$ in $\text{Int } K$) on finite samples of rational unit vectors, and confirms $\mu(A_K(x)) \geq 1$ at sampled points of each boundary class. It establishes none of the universal statements on which the theorem rests — the criteria at *every* boundary point, $\text{ext}(K) = C_1 \cup C_2$, the constraint $\mu(A_K(x)) \geq 1$ for *every* x , and $\text{ill}^*(B_1(\mathbb{C}^n)) = 2n$ — it hard-codes the cone mass $\mu(V_1) \geq 2$ of the lower bound rather than deriving it, and its second route to the non-zonoid conclusion imports [1]’s zonoid face lemma. Those steps are the closed-form arguments printed above; the program is not an independent verification of the theorem.

REFERENCES

- [1] L. Rotem, B. Schejter and B. A. Slomka, *The complex illumination problem*, Combinatorica **46** (2026), no. 1, Paper No. 3. doi:10.1007/s00493-025-00195-7; e-print arXiv:2410.12021.
- [2] M. Naszódi, *Fractional illumination of convex bodies*, Contrib. Discrete Math. **4** (2009), no. 2, 83–88.
- [3] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd ed., Encyclopedia of Mathematics and its Applications **151**, Cambridge University Press, 2013.